

Bike Sharing Demand Analysis Using Excel

1. Executive Summary

This project analyses **1,000 hourly bike-sharing records** from **1 January to 14 February 2011** using Microsoft Excel. The analysis covers data preparation, demand patterns, rider behaviour, calendar and weather relationships, forecasting, anomaly detection, and dashboard reporting.

Metric	Value
Total rentals	58,304
Average rentals per hour	58.30
Registered / Casual	91.6% / 8.4%
Peak hour — 17:00	140.20
Working / Non-working day	64.14 / 47.65
Clear / Light rain	61.67 / 33.20
First 7-day forecast	1,500.49/day

Key Takeaways

- Demand is highest at **08:00 and 17:00**, with registered riders showing the strongest peak-hour pattern.
- **Registered riders account for 91.6%** of total rentals.
- Working-day demand is higher than non-working-day demand.
- Light-rain demand is approximately **46% lower** than clear-weather demand.
- Temperature has a stronger association with **casual** than registered rentals.

Business implication: Bike availability and rebalancing should be prioritized around peak periods, while weather conditions and short-term forecasts can be used as supporting planning inputs.

2. Objective, Dataset & Methodology

Objective

The analysis examines how **time, rider type, calendar factors, and weather** relate to rental demand and converts the observed patterns into practical planning insights.

Dataset Summary

Item	Detail
Records / period	1,000 hourly records, 1 Jan – 14 Feb 2011 (45 days)
Rentals per hour	Average 58.30; minimum 1; maximum 249; standard deviation 50.99
Registered / casual	53,383 / 4,921
Season	Winter only

Data note:

Hours with no rentals are not represented as rows. The final date, **14 February 2011**, contains only 8 hourly observations and ends at 07:00.

Methodology

Source Data → Power Query Cleaning → Merge & Append → Lookup Enrichment → PivotTables & Charts → Correlation Analysis → Forecasting → Anomaly Detection → Dashboard

Data preparation included data-type cleaning, validation, duplicate checks, lookup-based enrichment, and replacement of **11 missing values using the mean**

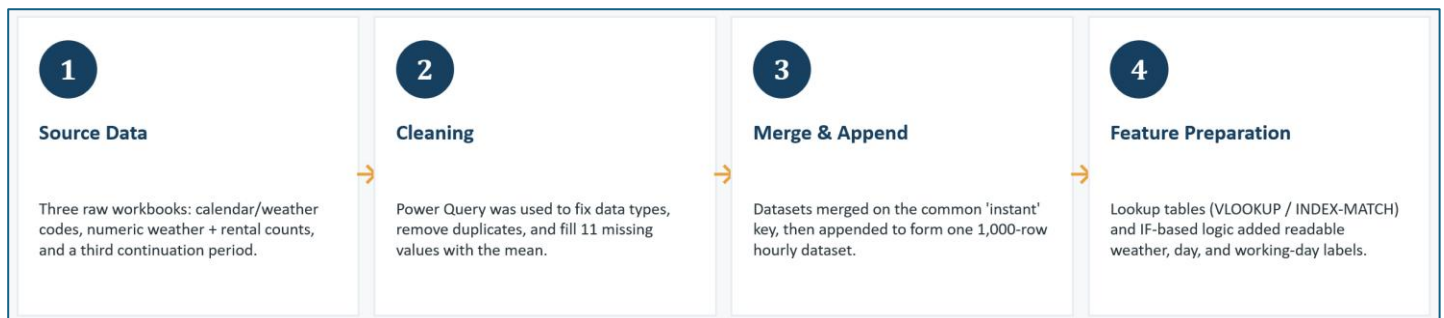


Figure 1. Data preparation workflow

3. Demand & Rider Analysis

Demand is highest during the **morning and evening peak periods**, while early-morning demand is substantially lower.

Hour	Average rentals
17:00	140.20
08:00	128.29
04:00	2.16

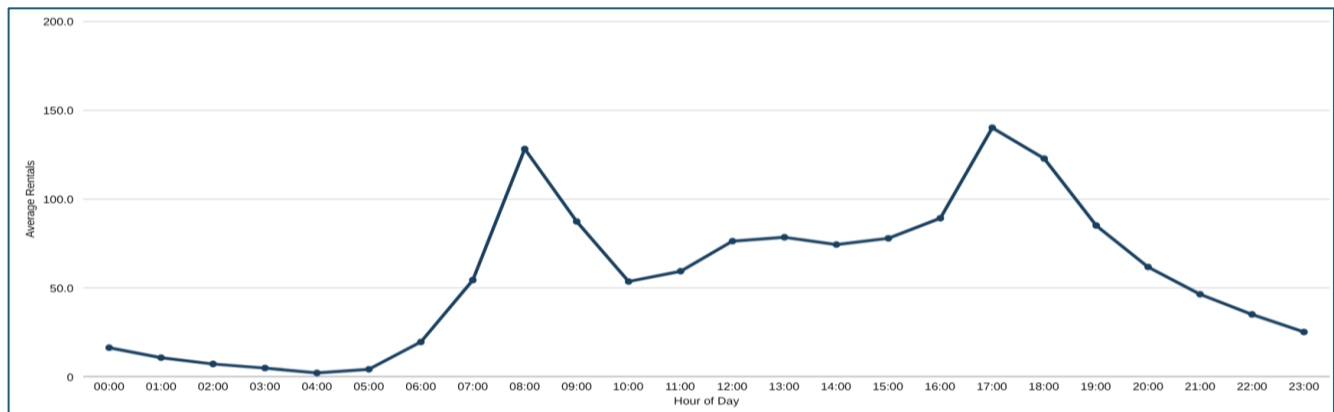


Figure 2. Average rentals by hour

Rider Type Pattern

Registered riders account for **53,383 rentals (91.6%)**, compared with **4,921 casual rentals (8.4%)**.

Registered riders show stronger morning and evening peaks, while casual demand is more distributed across daytime hours. Registered demand reaches **131.23 rentals at 17:00**, while casual demand peaks at **12.47 rentals at 15:00**.

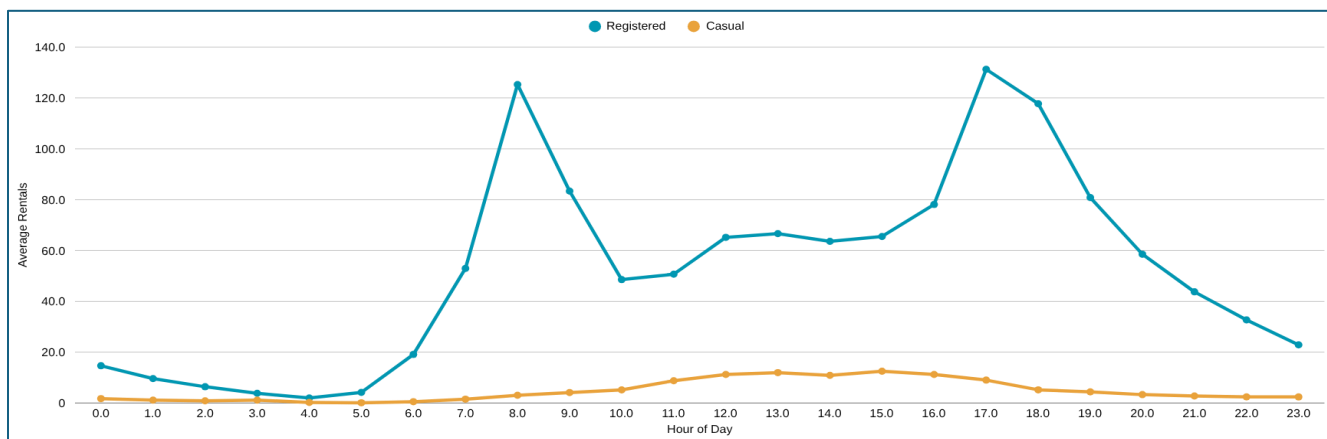


Figure 3. Registered vs casual by hour

4. Calendar Analysis

Demand is higher on **weekdays and working days** than on weekends and non-working days.

Comparison	Higher Average	Lower Average
Weekday vs weekend	63.34	48.09
Working vs non-working day	64.14	47.65

Thursday is the highest day of the week (**67.66**) and Saturday the lowest (**46.67**).

Holiday records average **41.67 rentals per hour**, compared with **58.71** for non-holiday records. However, all 24 holiday records come from **17 January 2011**, so this should not be interpreted as a general holiday effect.

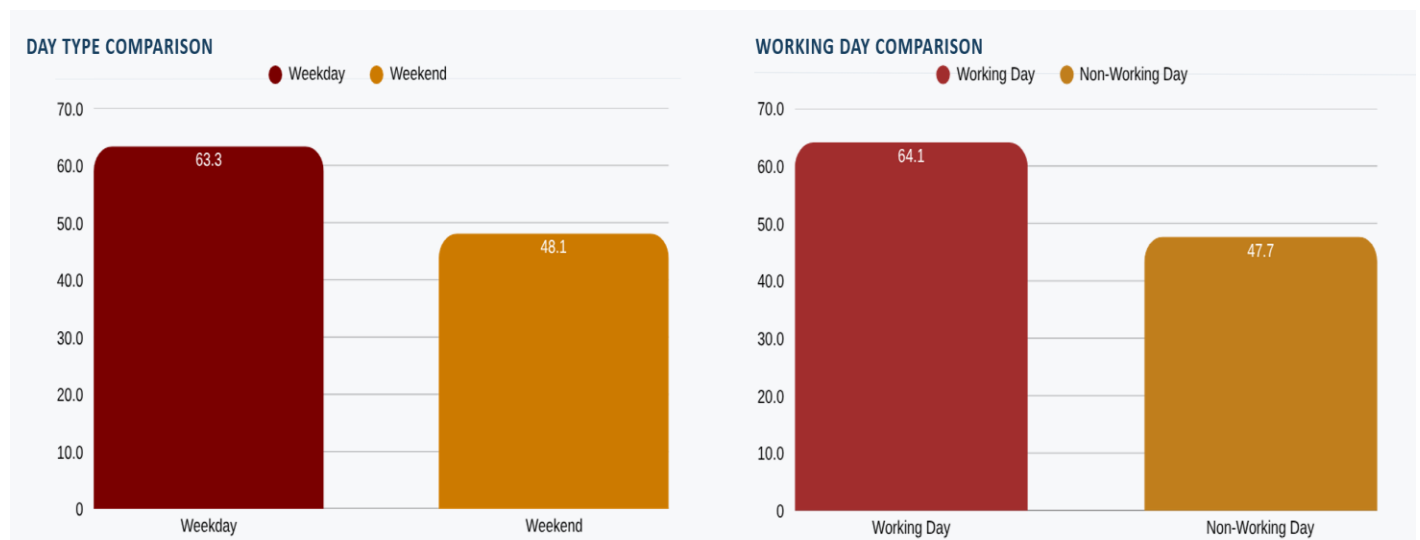


Figure 4. Weekday/weekend and working-day comparison.

5. Weather & Environmental Analysis

Light rain reduces average demand to about half of clear-weather levels (33.20 vs 61.67).

Weather	Records	Average rentals per hour
Clear	606	61.67
Cloudy	309	58.60
Light rain	84	33.20
Heavy rain	1	36.00 (not reliable)

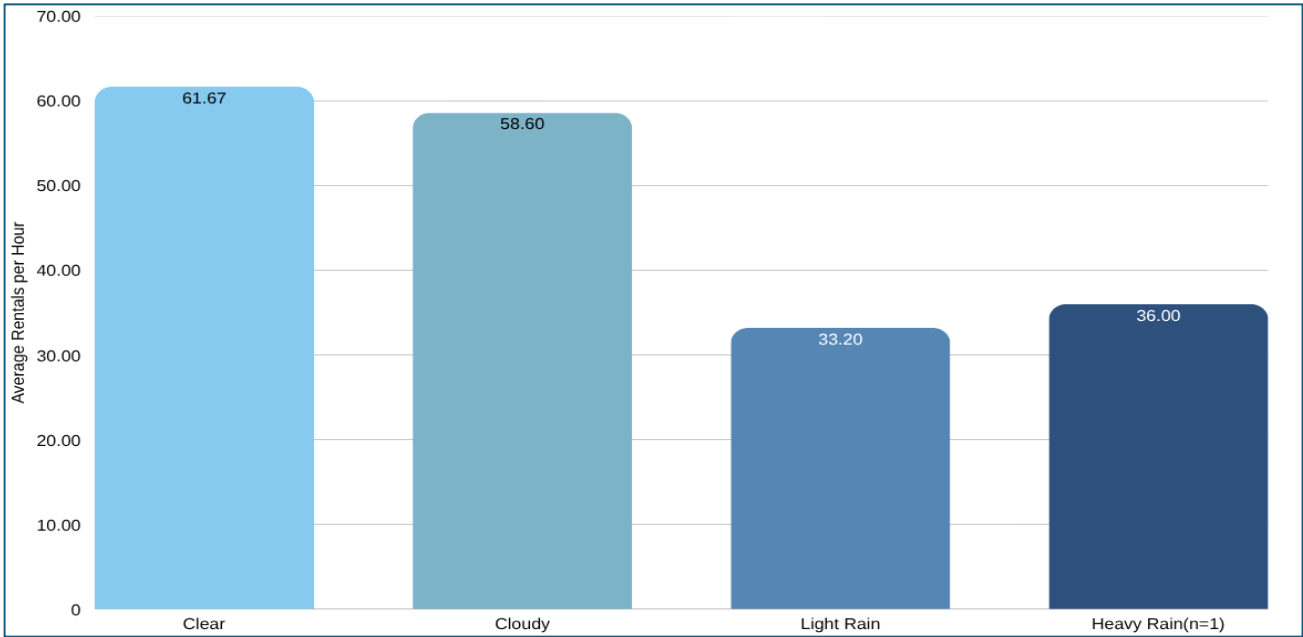


Figure 5. Average rentals by weather

Pearson correlation with rentals

Variable	Total	Casual	Registered
Temperature	+0.229	+0.470	+0.169
Apparent temperature	+0.193	+0.424	+0.138
Humidity	-0.254	-0.259	-0.229
Windspeed	+0.104	+0.129	+0.090

- Temperature has the strongest association with **casual** rentals.
- Humidity has a weak negative association with demand; windspeed is weak.
- Correlation shows association, not causation.

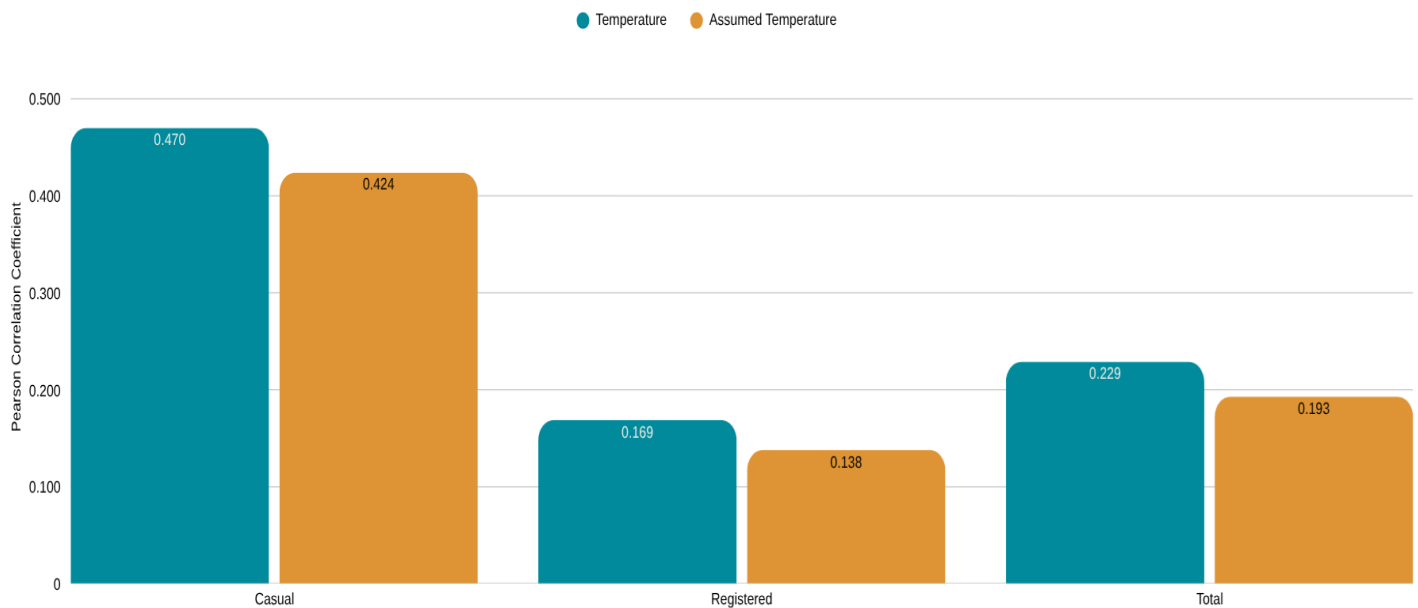


Figure 6. Correlation chart

6. Temperature & Forecast

Demand increases through the lower temperature bands and reaches the highest observed average in the **0.3–0.4 band at 93.88 rentals/hour**, before declining to **59.48** in the 0.4–0.5 band.

Temperature values are normalized from **0 to 1**, so these bands represent normalized temperature rather than direct temperature units.

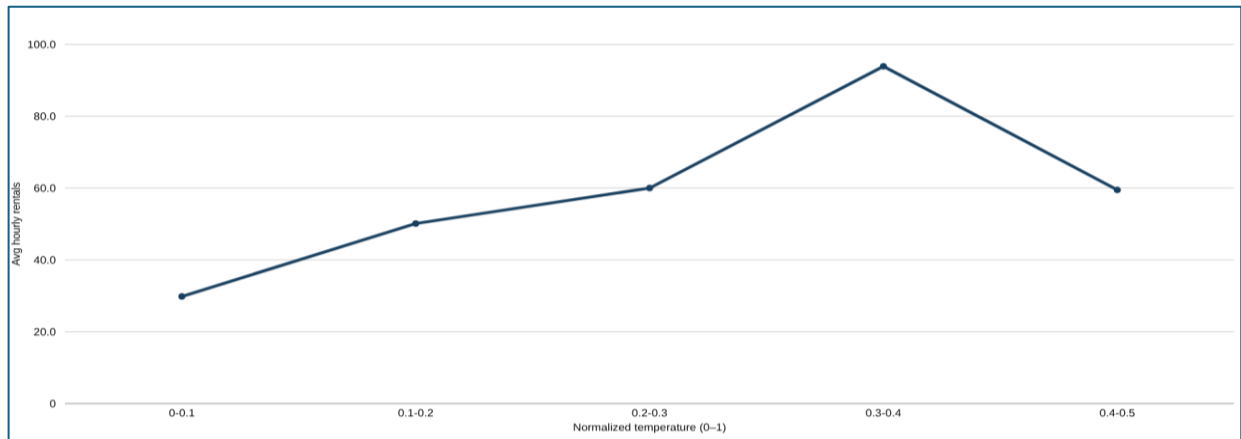


Figure 7. Demand by temperature band

Excel's FORECAST.ETS projected daily demand for 15 Feb – 7 Mar 2011 (21 days).

Forecast measure	Rentals per day
Historical average (45 days)	1,295.64
First 7 forecast days	1,500.49
Full 21 forecast days	1,543.28

Based on the Forecast method: Excel FORECAST.ETS, the model projects an average of about 1,500 rentals/day for the first 7 forecast days — a projection, not a guaranteed outcome. Prediction intervals are not shown.

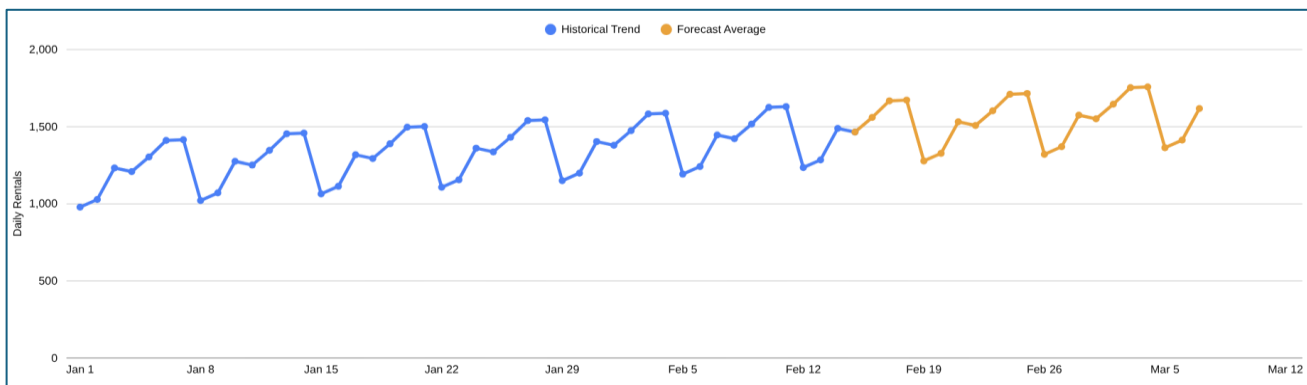


Figure 8. Historical and forecast demand

7. Anomaly Detection & Dashboard

The **IQR method** was used to identify unusually high or low daily rental totals.

Metric	Value
Q1	1,005
Q3	1,562
IQR	557
Lower Threshold	169.5
Upper Threshold	2,397.5

One of the 45 days was flagged: **14 February 2011**(151 rentals).

However, only **8 hourly observations** are available for that date, ending at 07:00. Therefore, the result is best treated as a **partial-day/data-coverage issue rather than a confirmed demand anomaly**.

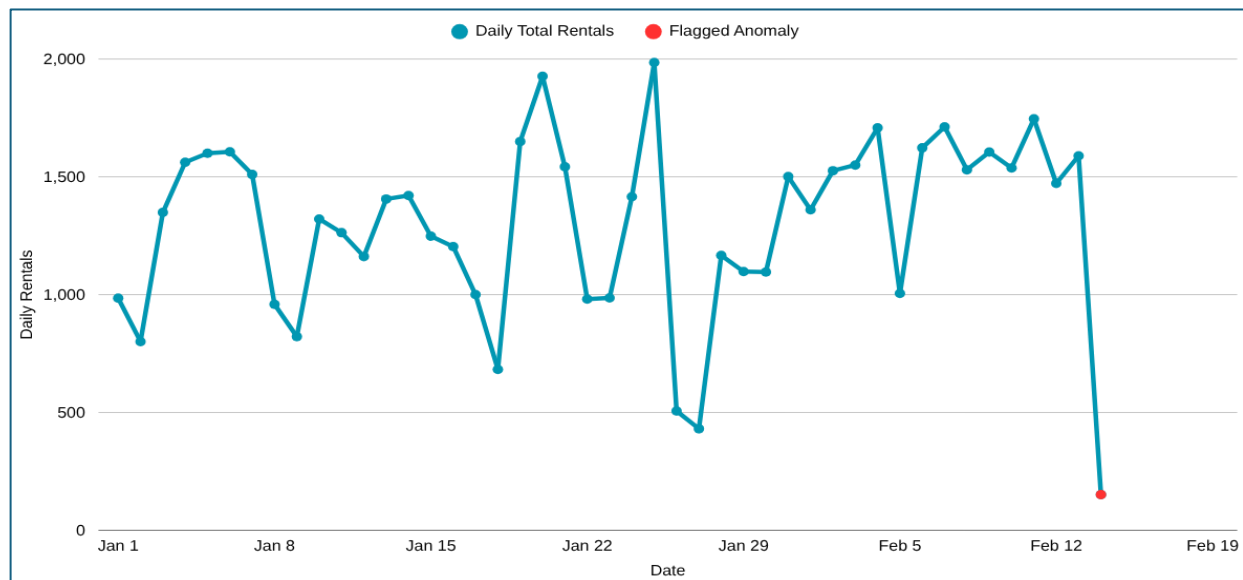


Figure 9. Daily rentals with flagged anomaly

8. Excel Dashboard

The dashboard combines KPIs, hourly, rider, weather and forecast charts, with slicers and a timeline for filtering. Power Query and PivotTables provide the underlying **refreshable reporting workflow**.



Figure 10. Excel Dashboard

9. Findings, Recommendations & Conclusion

Key Finding	Recommendation
Registered riders account for 91.6% of rentals and show the strongest 08:00 and 17:00 peaks.	Prioritize bike availability and rebalancing before peak periods.
Weekend demand is about 24% lower than weekday demand.	Adjust capacity and operational planning by day type.
Light-rain demand is about 46% lower than clear-weather demand.	Use weather forecasts as a supporting planning input.
Temperature has a stronger association with casual riders.	Consider weather conditions when planning casual-rider capacity.
First 7-day forecast averages about 1,500 rentals/day.	Use the forecast for short-term planning and refresh it with new data.
Only 45 winter days are available.	Collect longer, multi-season data to confirm seasonal patterns.

Limitations

- Winter-only data covering 45 days; 14 Feb is a partial day.
- Heavy rain has one observation; the holiday result comes from one holiday.
- Correlation does not establish causation.
- The forecast rests on a short history and is indicative only.

Conclusion

Bike rental demand in this dataset varies mainly by **time of day, working-day status, rider type, and weather**. Registered riders dominate total demand and show the strongest peak-hour pattern, while casual rentals have a stronger association with temperature.

The Excel workflow provides an end-to-end solution covering **data preparation, integration, analysis, forecasting, anomaly detection, and dashboard reporting**. These results provide a practical basis for bike availability and short-term planning, within the limits of the available winter-period data.